

Instructor: Date:

Learner: Regional Branch:

Program: Center:

CONTENT

- Let's Start! • Let's Explore! • Let's Practice! • Extension Activity • Target Words • Learning Points



Let's Start!

Data and Databases



Learning Outcome:

Identify key information about people's jobs and places of work in spoken or written contexts.

Design the data structure, starting from the conceptual model determined in the systems analysis, using database IT tools, according to regulation and established standards.

Can Do Statements:

- I can explain the data resources in information systems.
- I can give examples of data resources in information systems.
- I can analyze information related to data resources in information systems.

Ask your partner:

- What do you think the main purpose of a database is? Explain.
- What kind of data do you think one can save in a database? Why?





Let's Explore!

A professor and a student are exchanging ideas on data resources in information systems.



Instructions: Read the text below.

Student: Professor, I still don't understand the concept of data resources in an information system. Could you explain to us what it is, please?

Professor: Certainly. It's the collection of raw material data from the information system and, of course, a database that will store meaningful and purposeful data. Now, a database is the composition of several resources, such as bits, bytes, fields, records and files. A database is, then, a series of files; this chart, here, illustrates my point. A field resource can be either a word, an audio or a number. So, along your career, in the future, you'll find databases comprising numbers, texts, alphanumeric data, images, videos, and audios. Tomorrow you'll practice some more making a database, OK?

Student: Sure, but Could you give us some examples?

Professor: All right. Let's imagine you are collecting your classmates' personal data, so each piece of raw material data, like their names, last names, and mobile numbers, is what we will define as a field. Similarly, The collection of fields you get from every student is what we will call the record. Once you finish collecting all the required data from every student in the records, you'll soon have a file. Now, taking into account what I told you before, if you collect files from several students of your class, you will soon have a database, that is, a collection of files.

Student: Thanks professor.

Professor: Finally, we create the database with a specific purpose, that is, to store and analyze the information. Also, bear in mind that to do so, you'll have to design the database, including all the required information without creating redundant fields. Besides, it's important to design an appropriate structure to ease calculations and retrieve information easily.

Student: Thanks professor, it's been really helpful.

**Let's Practice!**

Let's talk about the data resources you need to bear in mind.

Instructions: Select the correct answer.

1. Data will _____ meaningful information after you properly _____ it.
 - a. produce, combine
 - b. produce, will combine
 - c. produce, combines
 - d. produce, combined
2. Later, I _____ you this chart to illustrate a data base is composed of bits, bytes, fields, records, and files.
 - a. show
 - b. showed
 - c. shows
 - d. will show
3. After you start your career, you will _____ that a database consists of numbers, text, alphanumeric data, images, videos, and audios.
 - a. seek
 - b. come
 - c. find
 - d. finds

**Let's Practice!**

Every year the company has to update its databases.

Instructions: Organize the mystery sentence.

1. information. / company / to / analyze / and / databases / creates / store / Our

2. the / raw / You / with / all / a / to / data. / database / design / need / material

3. facilitate / designs / company / My / appropriate / structures / calculations. / that



Extension Activity

How to create a database?

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Search for information about data resources in information systems.
- Identify how a database works.
- Make a list of steps you will follow to create a database. Give specific examples (e.g. when a gym needs to gather information about their customers...)
- Share your list of steps with your class and discuss.

Self-Assessment

Yes

No

- I search for information about data resources in information systems.
- I identify how a database works.
- I make a list of steps I will follow to create a database.
- I share information and discuss about the topic.

☐ ☐
☐ ☐
☐ ☐
☐ ☐


Target Words

- meaningful** (adjective): showing or having meaning, significance or value
e.g. It's the collection of raw material data from the information system and, of course, a database that will store meaningful and purposeful data.
- purposeful** (adjective): showing the purpose of what you are doing
e.g. It's the collection of raw material data from the information system and, of course, a database that will store meaningful and purposeful data.
- record** (noun): the collection of facts about someone or something in a document or in a digital file
e.g. The collection of fields you get from every student is what we will call the record.
- bear in mind** (idiom): to remember some information that is important before making a decision or think about something
e.g. Also, bear in mind that before you can do that you'll have to design a database.
- raw material data** (noun): the basic information you need to make a database
e.g. Let's imagine you are collecting your classmates' personal data, so each piece of raw material data, like their names, last names, and mobile numbers, is what we will define as a field.



Learning Points

Future with Will in Affirmative and Negative with Time Expressions

Use the modal auxiliary **will** to express ideas in future time.

When you use the modal auxiliary **will** to express future ideas, you have to take into account the following functions and applications of it:

Functions

1. To make a quick decision at the time of speaking.
2. To make a prediction about the future based on non-present or factual evidence.
3. To make a promise, an offer or threat.
4. To make a prediction based on a habit or predictable behavior.

In order to make a sentence in future simple with will, you need to use the following structure:

Affirmative: **Subject + will + verb + complement.**

Negative: **Subject + will + not (won't) + verb + complement.**

Note that in the negative for, the contraction of **will not** is **won't**.

Examples:

- *Now, that I have all the information, **I'll create** a database with it.*
- *Samantha **won't collect** information because she already did it last time.*

Time expressions with Future will

- There are many time expressions that you can use with the future simple with will. They can be used in affirmative, negative and interrogative constructions.
- The following is a short list of options:
- Tonight
- Tomorrow, tomorrow morning, tomorrow afternoon, etc.
- The day after tomorrow
- Next week, next month, next year, etc.
- In an hour, in a day, in a week, in a month, in a year, in June, etc.
- This week, this month, this year, this century, etc.
- On Monday, on my birthday, on Sunday, on 13th March 2025, etc.
- Soon, after, later, in the near future, in two or three days, in a little while, in a few minutes, etc.

Examples:

- *The boss **will call later** to see if we finished the database.*
- ***I won't be able** to collect all the users information to create the database **by tomorrow morning**.*





Learning Points

Examples

- *The collection of fields you get from every student is what we will call the "record."*
- *So, along your career, in the future, you'll find databases comprising numbers, texts, alphanumeric data, images, videos, and audios.*

Computer Metaphors

There are some every day words used in the world of computers that over the years have popularized more and more.

Space: It usually refers to the capacity of storing information a computer has.

Port: a slot where you can connect different kinds of devices so that you can share information or allow a computer to export information.

Tools: Like mechanical tools, a program is full of virtual tools capable of performing tasks that help to solve problems, failures and inconsistencies.

Cut: a command you use either to trim images and texts from a document or to change a file, document or folder from one specific place in the hard drive to another.

Clipboard: Virtual space where most copied and cut objects are placed temporarily for later use.

